J-C-30A
AMENDMENT-1
December 9, 1974

FEDERAL SPECIFICATION

CABLE AND WIRE ELECTRICAL
(POWER, FIXED INSTALLATION)

This amendment which forms a part of Federal Specification J-C-30A, dated October 7, 1970, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

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Paragraph 1.1, second line, after "aluminum" add "or copper clad aluminum."

Change paragraph No. from 1.3 to 1.2.2.

Delete paragraph 1.2.2 in its entirety and substitute as follows:

1.2.2 Type designation. Cables and wires covered by this specification shall be identified by a type designation, formed as indicated below. This type designation is intended for cataloging and ordering purposes, and not for surface printing on the wire or cable.

J-C-30A	RHW	10	A	E	4/14	SR	SJ	G	9
Spec. No. (1.2.3)	UL Type (1.2.4)	Maximum working voltage (1.2.5)	Conductor material (1.2.6)	Type of conductors (1.2.7)	No. and size of conductors (1.2.8)	Type of insulation (1.2.9)	Outer covering (1.2.10)	Ground wire (1.2.11)	Color code of single conductor (1.2.12)

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Table I, delete in its entirety and substitute:

TABLE I. Cable and wire types

UL Types	UL Standard No.	Description	Maximum Working Voltage	Temp. °C.
RHW	44	Heat and Moisture Resistant	5000*	75
RHH	44	Heat resistant	600	90
***RHH or RHW	44	Heat and moisture resistant	600	90
SA	44	Silicone, heat resistant	600	90
TW	83	Moisture resistant	600	60
THWN	83	Heat and Moisture resistant	600	75
THW	83	Heat and Moisture resistant	600	75
THHN	83	Heat resistant, for use in dry locations	600	90
***THHN or THWN	83	Heat and moisture resistant	600	90
NM	719	For use in dry locations	600	60
NMC	719	For use in dry or moist and mildy corrosive locations	600	60
UF	493	For direct burial in earth	600	60
V	133	Dry locations only, general purpose	5000*	85
VD	133	Dry locations only, general purpose	5000*	85
VM	133	Dry locations only, general purpose	5000*	85

FSC 6145

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TABLE I. Cable and wire types (con.)

UL Types	UL Standard No.	Description	Maximum Working Voltage	Temp. °C.
AVA	115	Dry locations only, general purpose	5000*	110
AVB	115	Dry locations only, general purpose	5000*	90
AVL	115	Dry and wet locations, general purpose	5000*	110
A	115	Dry locations only, general purpose	300	200
AA	115	Dry locations only, general purpose	300	200
AI	115	Dry locations only, general purpose	300	125
AIA	115	Dry locations only, general purpose	600	125
SE	854	Flame and moisture retardant, no mechanical protection	600	60 or 75
USE	854	Moisture resistant, no mechanical protection, for underground use	600	60 or 75
***RHW or USE	44	Moisture resistant, for underground use	600	60 or 75
ACHH	4	Heat resistant, armored	600	90
ACT	4	Dry locations, armored	600	60
ACHL	4	Heat resistant, armored, lead sheath	600	75
XHHW	**	Heat and moisture resistant (cross linked polythylene, thin wall)	600	75 or 90

*Voltage rating is maximum value. Desired voltage rating should be specified (see 1.2.4).
**It is anticipated that this type will be included in UL 44.
***For dual marking, see paragraph 3.4.

Add new paragraph 1.2.3: 1.2.3 Cable and wire specification no. J-C-30A.

Change from 1.3.1 to 1.2.4.
from 1.3.2 to 1.2.5.

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Change from 1.3.3 to 1.2.6.
from 1.3.4 to 1.2.7.
from 1.3.5 to 1.2.8.

Add new material to 1.2.6 as follows:

CA - Copper clad aluminum (UL 83)

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Change from 1.3.6 to 1.2.9.
from 1.3.7 to 1.2.10.
from 1.3.8 to 1.2.11.
from 1.3.9 to 1.2.12.

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Change from 1.3.10 to 1.2.13.

Paragraph 1.2.13, third line change from 1.3 to 1.2.2 and 1.3.9 to 1.2.12: fifth line change 1.3.5 to 1.2.8.

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Paragraph 3.1, third line, delete 1.3.

Paragraph 3.3.1, fourth line, delete 1.3.9 and substitute 1.2.11.

Add new dual marking paragraphs as follows:

3.4 Dual marking.

3.4.1 THHN or THWN wire. Any Types of THHN wire that complies with the requirements in UL Standard 83 for Type THWN wire as well as those for Type THHN may be dual marked "THHN or THWN" if the manufacturer so desires.

3.4.2 RHH or RHW wire. A wire or cable which complies with the requirements in UL Standard 44 for Type RHH wire and all the requirements for Type RHW wire may be dual marked "RHH or RHW".

3.4.3 RHW or USE cable. If a rubber insulated wire or cable also qualifies completely for use as a cable in a different classification (such as Type USE service-entrance cable), the product may carry an amplified marking including the word "or" to indicate the additional classification and the optional use - for example, "RHW or USE".

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Paragraph 5.1.1.1, add to the end of the paragraph "Alternatively, the reels may be packaged in a round, form-fitting, shrunk wrapped polyethylene package. The film shall have a minimum thickness of 0.006 inches. A top hole shall be provided to dispense the wire.

Paragraph 5.1.2, add to the last sentence, "and sealing of the box is not required".

Paragraph 5.2.1.1, first line, delete "cord" and substitute "wire".

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Paragraph 5.2.2 delete in its entirety and substitute as follows:

5.2.2 Coils packaged in shrink wrapped polyethylene film as specified in 5.1.1.1, shall be packed four packaged coils to a box. The box shall conform to PPP-B-636, class domestic. All other coils and reels packaged as specified herein, for level A or B, shall require no packing.

Paragraph 6.1 (b), delete (1.3) and substitute (1.2.2).

Paragraph 6.1, add, (i) Whether civil or military marking is required (see 5.3).

MILITARY INTERESTS:

Custodians:

Army - EL
Navy - YD
Air Force - 17

Review activities:

Army - ME
Navy - AS, YD
Air Force - 80
NSA
DSA - IS

User activities:

Army - MI, MU
Navy - MC
Air Force - 11

Preparing Activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS, PBO
DOT-ACO
DC GOVT-DCG
NASA-JFK
HEW-FEC

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